Paxos Made Live Response Xiyao Long

The article is a description about the implement of Paxos system. More specifically it concerns about implementing a fault tolerant message sharing protocol, logging strategy and databases suitable for enabling the system.

The authors introduces the Paxos algorithm pretty well. The protocol enable elections of coordinators to disseminate a relatively current value of some data to other members. But the process is rather concurrent as every replica send and receive messages and are free to propose to be coordinator. The demand for a to-be-coordinator to receive “promise” from most of others and the consensus agreement of a data value ensure that the data the coordinator broadcasts is usually up-to-date.

The paper talked about their major contribution of implementing a very fault-tolerant system by let replicas each keep a copy of the data operation log locally. After system crash, Paxos can reconstruct its state from the logs, or in another situation, the lagged replicas can use the logs with newer operations to replay and update itself. Moreover, they also solved a major culprit of this approach. The everprowing logs can require indefinite disk spaces and prolong the state reconstruction time. They enable the system to let the client application to save a snapshot of the data structure of a data replica, the local log prior to the snapshot can then be discarded. The lagging-replica can also receive and install a snapshot instead of replaying a long sequence of replication.

I find the system implementation in this article did solve many problems in a large distributed system. It first solves the data value agreement problem, the recovering of state during system crashing and the lack of disk space imposed by logging. I think the use snapshot in their implementation can be used in general when constant logging is proposed but demands too much space. The specification for logging the a sequence of five messages also gives a complete description of the data operation and can be used for logging the process of information dissemination in general.